An explicit contradiction in Gamma makes Gamma inconsistent

The Open Logic Project contributors

openlogic:first-order-logic/sequent-calculus:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-sequent-calculus/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof uses the Left-exchange structural rule explicitly; rule labels render as operator letters under the upstream naming macros

Editorial note. The label prop:explicit-inc names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Negation, exchange and cut. Left-not turns A sequent A into (not A), A sequent (empty); Left-exchange reorders to A, (not A) sequent (empty); cut with pi closes the contradiction.. The displayed derivation chains Left-not, Left-exchange and cut after the derivation pi of Gamma_0 sequent A.

Proposition 0.1. *If $\Gamma \vdash A$ and $\neg A \in \Gamma$, then Γ is inconsistent.*

Proof. Suppose $\Gamma \vdash A$ and $\neg A \in \Gamma$. Then there is derivation π of a sequent $\Gamma_0 \Rightarrow A$. The sequent $\neg A, \Gamma_0 \Rightarrow$ is also derivable:

$$\dfrac{\begin{array}{c}\vdots\\ \pi\\ \vdots\\ \Gamma_0 A\end{array} \qquad \dfrac{\dfrac{AA}{\neg A, A}}{A, \neg A}}{\Gamma_0, \neg A}\ \text{Cut}$$

Since $\neg A \in \Gamma$ and $\Gamma_0 \subseteq \Gamma$, this shows that Γ is inconsistent. □